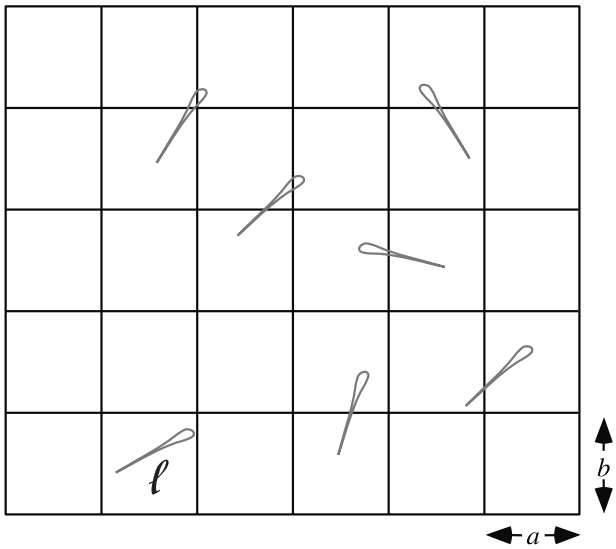
Buffon-Laplace Needle Problem



The Buffon-Laplace needle problem asks to find the probability $P(\ell, a, b)$ that a needle of length ℓ will land on at least one line, given a floor with a grid of equally spaced parallel lines distances a and b apart, with $\ell < a, b$. The position of the needle can be specified with points (x, y) and its orientation with coordinate ϕ. By symmetry, we can consider a single rectangle of the grid, so $0 < x < a$ and $0 < y < b$. In addition, since opposite orientations are equivalent, we can take $-\pi/2 < \phi < \pi/2$.

The probability is given by

$$P(\ell; a, b) = 1 - \frac{\int_{-\pi/2}^{\pi/2} F(\phi)\, d\phi}{\pi a b}, \tag{1}$$

where

$$F(\phi) = a b - b \ell \cos\phi - \ell a |\sin\phi| + \tfrac{1}{2} \ell^2 |\sin(2\phi)| \tag{2}$$

(Uspensky 1937, p. 256; Solomon 1978, p. 4), giving

$$2\ell(a+b) - \ell^2$$

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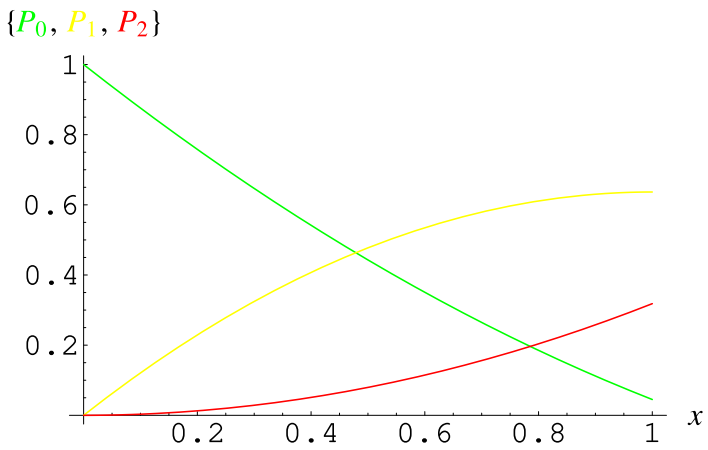

If $a = b$ so that $x = \ell / a = \ell / b$ and $0 < x < 1$, then the probabilities of a needle crossing 0, 1, and 2 lines are

$$P_0 = 1 - \frac{x(4-x)}{\pi} \tag{4}$$

$$P_1 = \frac{2x(2-x)}{\pi} \tag{5}$$

$$P_2 = \frac{x^2}{\pi}. \tag{6}$$

Defining N_i as the number of times in n tosses that a short needle crosses exactly n lines, the variable $N_1 + N_2$ has a binomial distribution with parameters n and m / π, where $m = x(4-x)$ (Perlman and Wichura 1975). A point estimator for $\theta = 1 / \pi$ is given by

$$\hat{\theta} = \frac{N_1 + N_2}{m n}, \tag{7}$$

which is a uniformly minimum variance unbiased estimator with variance

$$\operatorname{var}(\hat{\theta}) = \frac{\theta}{n}\left(\frac{1}{m} - \theta\right) \tag{8}$$

(Perlman and Wishura 1975). An estimator $\hat{\pi} = 1 / \hat{\theta}$ for π is then given by

$$\hat{\pi} = \frac{x(4x - x^2)}{1 - \frac{N_0}{n}}. \tag{9}$$

This has asymptotic variance

$$\operatorname{avar}(\hat{\pi}) = \frac{\pi^2 (4x - x^2 - \pi)}{n x (x-4)}, \tag{10}$$

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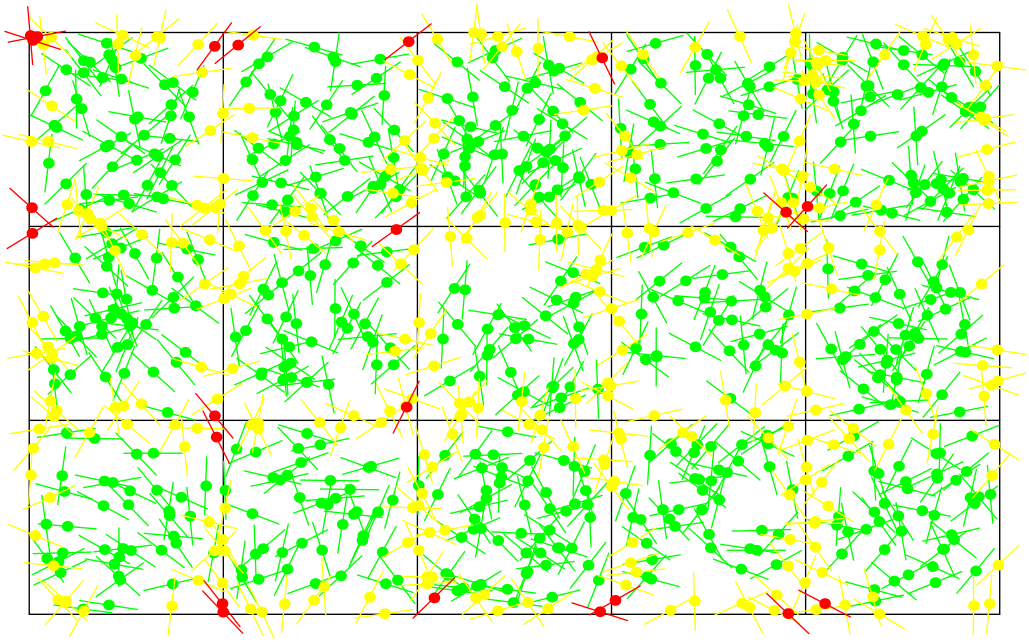
$$\approx \frac{0.465821}{n} \tag{12}$$

(OEIS A114602).



A set of sample trials is illustrated above for needles of length $a/\ell = b/\ell = 0.3$, where needles intersecting 0 lines are shown in green, those intersecting a single line are shown in yellow, and those intersecting two lines are shown in red.

If the plane is instead tiled with congruent triangles with sides a, b, c, and a needle with length ℓ less than the shortest altitude is thrown, the probability that the needle is contained entirely within one of the triangles is given by

$$P = 1 + \frac{(A a^2 + B b^2 + C c^2)\ell^2}{8\pi K^2} - \frac{(4a + 4b + 4c - 3\ell)\ell}{4\pi K}, \tag{13}$$

where A, B, and C are the angles opposite a, b, and c, respectively, and K is the area of the triangle. For a triangular grid consisting of equilateral triangles, this simplifies to

$$P = 1 + \frac{2}{3}\left(\frac{\ell}{a}\right)^2 - \frac{\ell\sqrt{3}}{\pi a}\left(4 - \frac{\ell}{a}\right) \tag{14}$$

(Markoff 1912, pp. 169-173; Uspensky 1937, p. 258).

SEE ALSO

Buffon's Needle Problem, Clean Tile Problem

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